

CAPE AGULHAS, South Africa

WMO: 689200

Lat: 34.8261S

Lon: 20.0131E

Elev: 13

StdP: 14.69

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.8	50.1	37.6	33.1	56.5	39.5	35.7	54.6	20.5	56.0	18.5	57.3	6.2	0	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	78.0	68.8	76.5	68.2	75.4	67.5	70.9	76.1	69.9	74.9	68.9	73.9	7.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.9	106.2	74.6	67.8	102.1	73.7	66.7	98.5	72.8	34.7	76.5	33.9	75.1	33.1	74.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.5	15.1	46.0	84.0	1.7	4.0	44.7	86.9	43.7	89.3	42.7	91.6	41.4	94.5		
(5)			41.9	72.6	1.2	1.4	41.1	73.6	40.4	74.4	39.8	75.2	38.9	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	70.5	70.2	68.4	64.8	61.8	58.5	57.9	58.0	59.7	62.5	64.8	68.5	(6)
(7)		DBStd	5.35	2.43	2.26	2.56	3.05	2.72	3.03	3.65	2.89	3.00	2.84	2.43	2.41	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	1067	0	1	3	37	105	197	224	218	162	86	32	2	(9)
(10)		CDD50	5028	636	566	571	445	365	255	245	249	291	387	444	574	(10)
(11)		CDD65	620	171	147	109	32	6	2	4	1	3	8	26	111	(11)
(12)		CDH74	627	231	153	75	31	6	9	15	6	15	4	2	80	(12)
(13)		CDH80	23	3	2	6	6	0	1	2	1	1	0	0	1	(13)
(14)	Wind	WSAvg	7.8	8.1	7.8	7.5	7.3	7.0	7.5	7.3	7.6	8.2	8.5	8.6	7.9	(14)
(15)	Precipitation	PrecAvg	17.7	0.9	0.7	1.1	1.9	2.0	2.2	2.4	1.9	1.2	1.5	1.1	0.7	(15)
(16)		PrecMax	23.1	5.0	2.5	5.4	10.0	5.7	5.5	5.5	5.1	2.8	4.4	3.6	4.1	(16)
(17)		PrecMin	10.3	0.0	0.1	0.2	0.4	0.5	0.7	0.8	0.6	0.3	0.2	0.2	0.1	(17)
(18)		PrecStd	3.3	1.2	0.5	1.2	1.9	1.0	1.1	1.2	1.0	0.6	1.2	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.0	79.1	77.9	78.0	74.2	72.8	75.5	70.2	76.2	74.0	74.0	78.0	(19)
(20)			MCWB	72.0	70.1	70.0	63.1	60.8	57.7	58.5	56.6	60.9	63.0	64.9	69.3	(20)
(21)		2%	DB	77.8	77.2	75.8	72.8	69.3	67.2	68.6	66.2	68.1	70.1	71.9	76.1	(21)
(22)			MCWB	70.2	69.0	68.5	64.6	60.0	57.2	56.3	56.8	59.7	63.2	63.9	68.2	(22)
(23)		5%	DB	76.4	75.9	74.3	70.8	67.4	64.7	65.1	64.2	66.0	68.5	70.7	74.7	(23)
(24)			MCWB	68.8	68.1	67.3	64.0	60.4	57.2	56.7	57.0	58.9	62.0	63.2	67.1	(24)
(25)		10%	DB	75.1	74.7	72.9	69.2	66.1	63.1	63.1	62.8	64.5	67.2	69.6	73.5	(25)
(26)			MCWB	67.7	67.3	66.1	63.3	60.0	56.8	56.4	56.7	58.0	61.0	62.4	66.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	71.6	70.9	67.9	63.9	61.2	61.0	60.9	63.4	66.1	66.8	71.0	(27)
(28)			MCD B	78.4	77.3	77.0	72.5	66.8	66.0	68.4	64.4	68.9	72.1	71.9	76.0	(28)
(29)		2%	WB	71.3	70.3	69.5	66.4	62.8	60.0	59.2	59.8	61.3	64.6	65.3	69.5	(29)
(30)			MCD B	76.2	75.4	74.5	70.6	66.4	64.1	64.7	63.3	66.8	69.0	70.2	74.6	(30)
(31)		5%	WB	70.1	69.3	68.4	65.3	62.0	59.0	58.2	58.9	60.1	63.4	64.3	68.3	(31)
(32)			MCD B	75.1	74.4	72.8	69.3	65.8	63.2	63.2	62.5	64.9	67.4	69.1	73.5	(32)
(33)		10%	WB	68.9	68.4	67.4	64.3	61.1	57.9	57.3	57.7	58.8	62.1	63.4	67.1	(33)
(34)			MCD B	74.0	73.5	71.7	68.2	65.1	62.1	62.0	61.7	63.5	66.1	68.3	72.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	8.2	8.2	8.0	8.2	8.9	9.3	9.2	8.9	8.0	8.5	8.3	(35)
(36)			MCDBR	9.4	9.5	9.8	9.9	10.0	11.4	12.8	11.2	10.6	9.2	9.7	9.6	(36)
(37)			MCWBR	5.4	5.5	5.6	5.7	6.0	7.2	7.4	6.6	6.2	5.5	5.8	5.8	(37)
(38)		5% WB	MCDBR	8.5	8.4	9.0	8.1	8.3	9.3	10.7	8.8	10.0	8.1	8.6	8.6	(38)
(39)			MCWBR	5.4	5.2	5.3	5.3	5.6	6.3	7.1	5.9	6.2	5.3	5.6	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.362	0.365	0.352	0.335	0.329	0.321	0.322	0.340	0.352	0.369	0.355	0.361	(40)	
(41)		taud	2.508	2.510	2.541	2.575	2.574	2.569	2.560	2.500	2.447	2.422	2.496	2.498	(41)	
(42)		Ebn at Noon	309	301	293	281	265	258	264	274	288	296	309	310	(42)	
(43)		Edh at Noon	36	35	31	27	23	22	23	28	34	37	36	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2381	2119	1708	1264	920	777	855	1106	1514	1909	2279	2453	(44)	
(45)		RadStd	81	82	89	57	49	40	38	50	71	109	89	106	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110

Nomenclature: See separate page